Law of Cosines

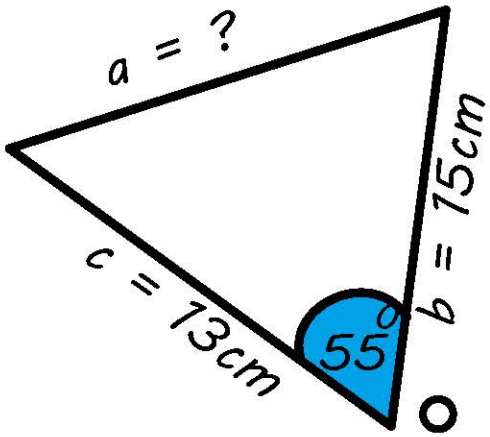
Finding sides

We can use the law of cosines when we know…

2 sides

1 angle Opposite the side we are trying to find.

For example…



We know 2 sides and the angle opposite the side we want to find, so we can use the cosine rule:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

For this example…

$a^2 = 15^2 + 13^2 - (2 \times ___ \times ___ \times \cos 55^\circ)$

$a^2 = 170{\cdot}305\cdots$

$a = \sqrt{170{\cdot}305\cdots}$

$a =$ _____ (to 1 d.p)

The law of cosines can be rearranged to make either a, b or c the subject, or the angles, cosA, cosB or cosC

$a^2 = b^2 + ___^2 - 2bc \cos A$

$b^2 = a^2 + c^2 - 2ac \cos ___$

$c^2 = ___^2 + b^2 - 2ab \cos C$

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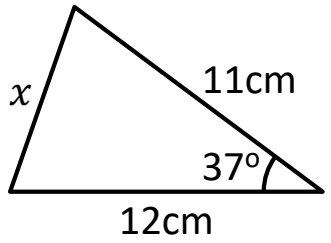
Q1 Work out the missing side, x, of each of the following triangles to one decimal place.

[a]



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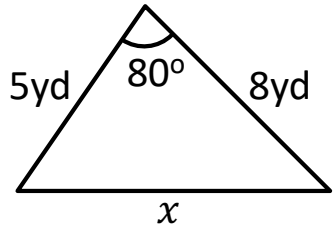

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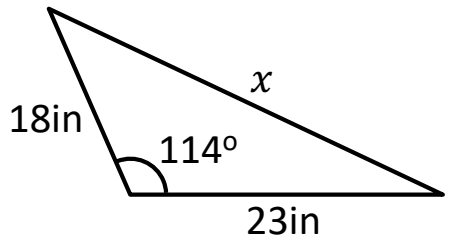

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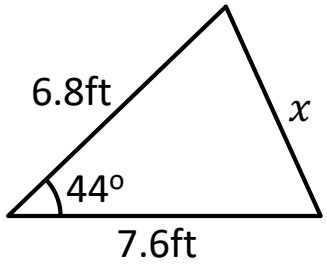

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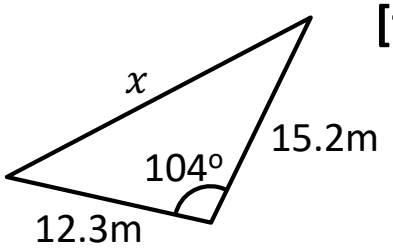

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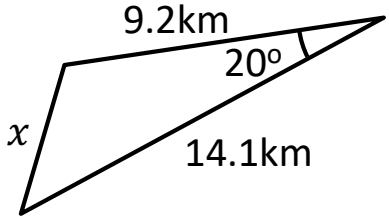
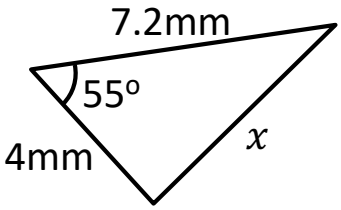
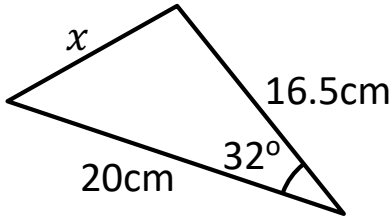


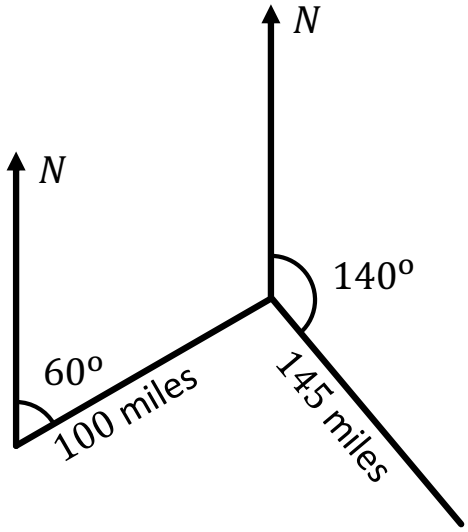
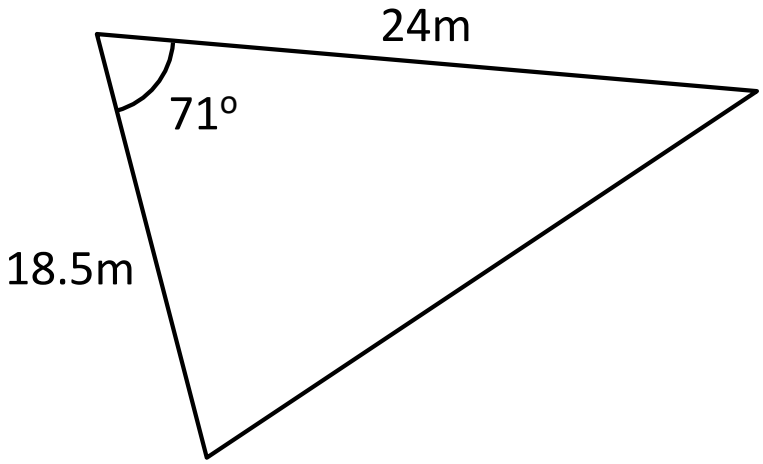
[g]



[h]



Q2 A ship sails from port A on a bearing of 060° for 100 miles and then on a bearing of 140° for 145 miles. Calculate the distance of the ship from port A to the nearest mile.



Q3 Find the perimeter of the following triangle.